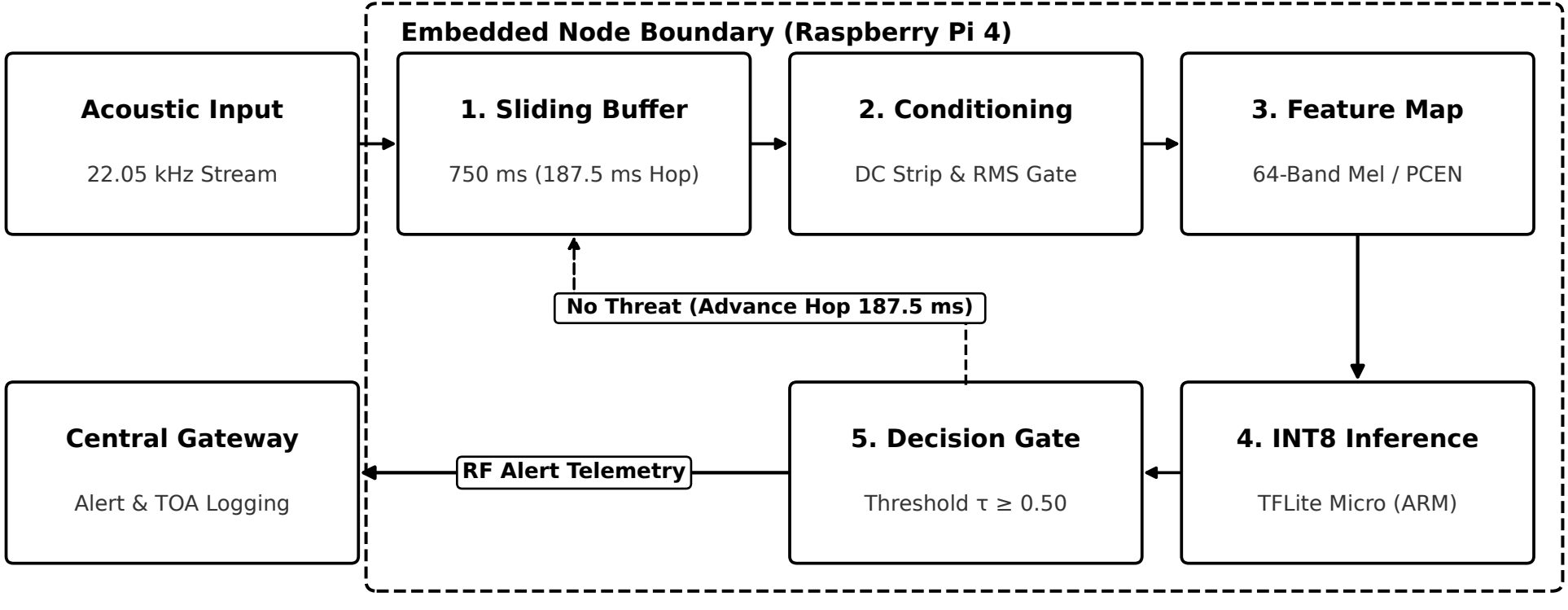
Embedded Node Boundary (Raspberry Pi 4)
Acoustic Input
22.05 kHz Stream
1. Sliding Buffer
750 ms (187.5 ms Hop)
2. Conditioning
DC Strip & RMS Gate
3. Feature Map
64-Band Mel / PCEN
No Threat (Advance Hop 187.5 ms)
Central Gateway
Alert & TOA Logging
RF Alert Telemetry
5. Decision Gate
Threshold τ ≥ 0.50
4. INT8 Inference
TFLite Micro (ARM)